

P1355R0

Exposing a narrow contract for ceil2

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This version:

<http://wg21.link/P1355R0>

Author:

[Chris Kennelly](#) (Google)

Audience:

LEWG, LWG, SG6

Project:

ISO/IEC JTC1/SC22/WG21 14882: Programming Language — C++

Abstract

ceil2 promises an unspecified value for out-of-bounds arguments. Out-of-bounds arguments should instead be undefined behavior.

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§ 1. Background

During the review of [\[P0556R1\]](#), LEWG requested `ceil2` return an unspecified value for out-of-bounds arguments. [\[P0556R3\]](#) was adopted in Rapperswil with this specification. [\[N4778\]](#) (24.6.4 "Integral powers of 2"):

Returns: The minimal value `y` such that `ispow2(y)` is `true` and `y >= x`; if `y` is not representable as a value of type `T`, the result is an unspecified value.

At the [\[SAN\]](#) meeting, LEWG took a poll reconsidering this decision at the request of SG6:

Change `ceil2` to hard UB, ill-formed in constexpr

SF	F	N	A	SA
11	5	1	0	0

As discussed in [\[P1233R0\]](#) (out-of-bounds `shift_left` / `shift_right`), silently accepting out-of-bounds values can hide bugs. Making this explicitly undefined behavior allows it to be detected and flagged by analysis tools.

§ 2. Proposal

Wording relative to [\[N4778\]](#):

```
template<class T>
    constexpr T ceil2(T x) noexcept;
```

- Expects: A value `y` representable as a value of type `T` exists where `ispow2(y)` is `true` and `y >= x`.
- Returns: The minimal value `y` such that `ispow2(y)` is `true` and `y >= x` ; ~~if `y` is not representable as a value of type `T`, the result is an unspecified value~~ .
- Remarks: This function shall not participate in overload resolution unless `T` is an unsigned integer type ([basic.fundamental]).

§ 3. Discussion

For values of `x` that fail to satisfy the *Expects* precondition, this results in undefined behavior (15.5.4.11).

§ References

§ Informative References

[N4778]

Working Draft, Standard for Programming Language C++. 2018-10-08. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/n4778.pdf>

[P0556R1]

P0556R1: Integral power-of-2 operations. 2017-03-19. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2017/p0556r1.html>

[P0556R3]

P0556R3: Integral power-of-2 operations. 2018-06-06. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p0556r3.html>

[P1233R0]

Shift-by-negative in `shift_left` and `shift_right`. 2018-10-02. URL: <http://www.open-std.org/jtc1/sc22/wg21/docs/papers/2018/p1233r0.pdf>

[SAN]

Meeting minutes for P0556. 2018-11-09. URL: <http://wiki.edg.com/bin/view/Wg21sandiego2018/P0556>